

Life Cycle Assessment (LCA): Smartphone

EXTENDED RESPONSE (QER)

Topic C6 / B6: Environmental Chemistry & Sustainable Resources

Candidate Name: _____ Class / Set: _____ Date: _____

1 Life cycle assessment of consumer electronics

A life cycle assessment assesses the environmental impact of products such as smartphones. Some activities in the life cycle of a smartphone are listed below.

- Extracting oil from oil wells
- Making plastics from oil
- Obtaining copper from copper ore
- Putting together all the components
- Packing the phone in a box
- Watching videos on the phone
- Taking apart the phone
- Recycling the copper from some components

Describe the four stages that are assessed in a life cycle assessment.

In your answer, give at least one example from the list above for each stage.



👉 Write your answer in continuous prose. Clearly identify each stage and link at least one activity from the list.

[6 Marks]

[illegible]

[Total: 6 marks]

Teacher Mark Scheme: 6-Mark Smartphone LCA

Assessment Objectives: AO1 (Demonstrate knowledge) & AO2 (Application to context)

Level (Marks)	Generic Description	Indicative Level Criteria
Level 3 (5–6)	A comprehensive and well-structured answer. All four stages of an LCA are correctly identified and clearly described. At least one correct activity from the list is matched to each of the four stages.	• 4 stages identified. • ≥ 1 valid example matched to each stage. • Coherent scientific terminology.
Level 3 (3–4)	A sound description of at least three stages with appropriate matching activities from the list, OR all four stages named with partially correct matching of activities.	• 3–4 stages named. • 2–3 stages correctly matched with activities. • Mostly organized structure.
Level 1 (1–2)	Identifies at least one or two stages of an LCA, or correctly assigns activities from the list to recognizable life cycle phases. Answer may be isolated or unstructured.	• 1–2 stages named or valid examples classified. • Basic recall without full synthesis.
0 marks	No creditworthy response or content completely irrelevant.	No relevant scientific content.

INDICATIVE SCIENTIFIC CONTENT (THE 4 LCA STAGES & EXAMPLES)

Stage 1: Extracting & Processing Raw Materials

Matching examples from list:

- *Extracting oil from oil wells* (crude oil for polymers/plastics).
- *Obtaining copper from copper ore* (mining, smelting, electrolysis).

Impacts: Mining ores, drilling oil, quarrying; causes habitat destruction, deforestation, toxic runoff, heavy machinery fossil fuel use.

Stage 2: Manufacturing & Packaging

Matching examples from list:

- *Making plastics from oil* (refining and cracking).
- *Putting together all the components* (assembling the smartphone).
- *Packing the phone in a box* (cardboard/package).

Impacts: High thermal and electrical energy consumption, industrial chemical emissions, wastewater, factory carbon emissions.

Stage 3: Use & Operation Over Lifetime

Matching example from list:

- *Watching videos on the phone* (charging and powering the device).

Impacts: Cumulative electricity drawn from the grid during recharging, battery life degradation, data-centre energy transmission.

Stage 4: Disposal & End of Life

Matching examples from list:

- *Taking apart the phone* (dismantling, sorting parts).
- *Recycling copper from phone components* (metal recovery).

Impacts: Landfill space, non-biodegradable electronic waste, heavy metals leaching into groundwater, toxic emissions if incinerated.

Examiner Guidance & Common Pitfalls:

- **Transport:** Transport and distribution occurs across *all* four stages (transporting crude oil, shipping parts, distributing to shops, hauling waste to landfill). *Do NOT accept transport as a distinct fifth stage.*
- **Recycling:** Recycling the copper is a disposal/reprocessing activity that mitigates the need to extract fresh raw materials in Stage 1.
- **Scoris Annotations:** Annotate with **L1, L2, L3** level descriptors and clear ticks on matched activity examples.